

Clauses in SQL

◆ What is Clauses:-

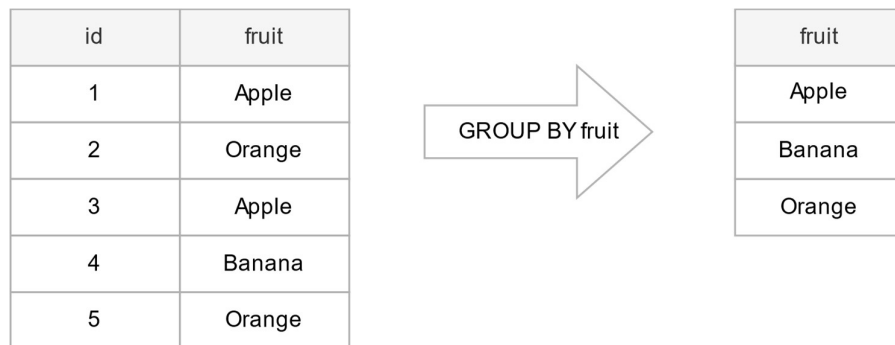
- SQL clauses are integral components of SQL statements that define **specific conditions** or **operations to be performed** on data within a database.
- They modify the behavior of SQL commands like **SELECT**, **INSERT**, **UPDATE**, and **DELETE** to filter, sort, group, or otherwise manipulate data efficiently.

◆ Common SQL clauses include:-

- **SELECT:** Specifies the columns to be retrieved from a table.
- **FROM:** Indicates the table(s) from which data will be retrieved.
- **WHERE:** Filters rows based on a specified condition.
- **GROUP BY:** Groups rows that have the same values in specified columns into summary rows.
- **HAVING:** Filters groups based on a specified condition, typically used with aggregate functions after GROUP BY.
- **ORDER BY:** Sorts the result set in ascending or descending order based on one or more columns.
- **LIMIT/TOP:** Restricts the number of rows returned by a query. (Syntax varies by database system, e.g., LIMIT in MySQL, TOP in SQL Server).
- **JOIN:** Combines rows from two or more tables based on a related column between them.
- **WITH (Common Table Expressions - CTEs):** Defines temporary, named result sets that can be referenced within a single SQL statement.

◆ GROUP BY Clause :-

- Arrange identical data into groups.
- The **GROUP BY** statement groups rows that have the same values into summary rows, like "find the number of customers in each country".
- The **GROUP BY** statement is often used with aggregate functions (**COUNT()**, **MAX()**, **MIN()**, **SUM()**, **AVG()**) to group the result-set by one or more columns.



➤ GROUP BY Syntax :-

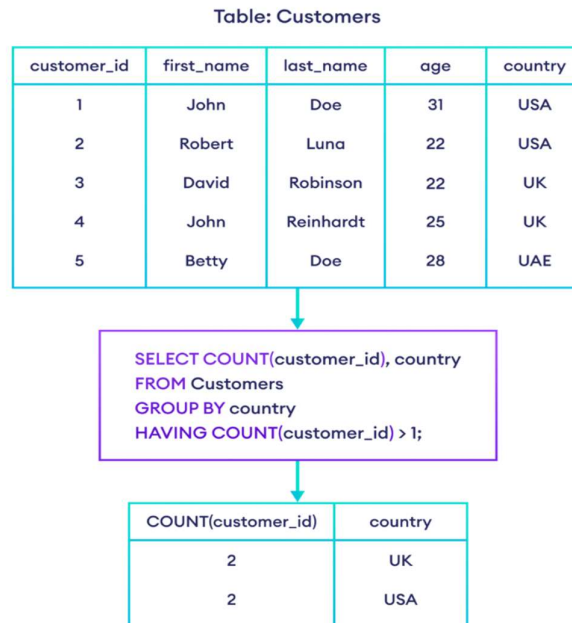
```
SELECT column_name(s)
FROM table_name
WHERE condition
GROUP BY column_name(s)
ORDER BY column_name(s);
```

➤ GROUP BY Example :-

```
SELECT COUNT(CustomerID), Country
FROM Customers
GROUP BY Country;
```

◆ **HAVING Clause:-**

- Filter records after the **GROUP BY** clause is applied.
- The **SQL HAVING** clause is used in conjunction with the **GROUP BY** clause to filter groups of rows based on aggregate functions.
- While the **WHERE** clause filters individual rows before grouping, the **HAVING** clause filters groups after aggregation.



➤ **HAVING Clause Syntax :-**

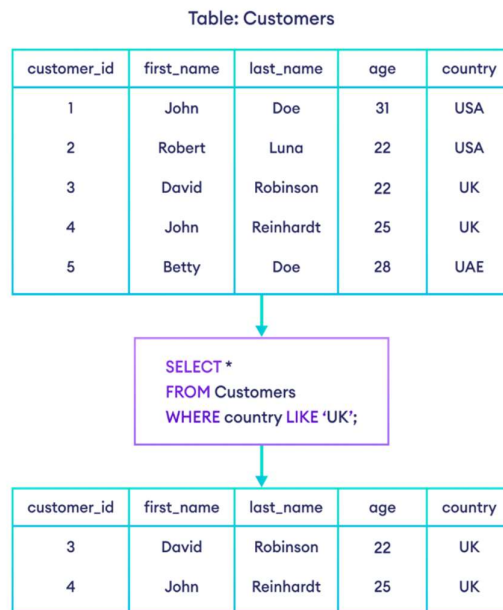
```
SELECT column_name(s)
FROM table_name
WHERE condition
GROUP BY column_name(s)
HAVING condition
ORDER BY column_name(s);
```

➤ **HAVING Clause Example:-**

```
SELECT COUNT(CustomerID), Country
FROM Customers
GROUP BY Country
HAVING COUNT(CustomerID) > 5;
```

◆ LIKE in SQL:-

- The **LIKE** operator in **SQL** is used in the **WHERE clause** to search for a specified pattern in a column. It allows for flexible pattern matching using **wildcard characters**.
- Wildcard Characters used with **LIKE**:-
 - **% :- (Percent sign)**: Represents zero, one, or multiple characters.
 - **_ :- (Underscore)**: Represents a single character.



➤ LIKE in Syntax:-

```
SELECT column1, column2, ...  
FROM table_name  
WHERE column LIKE pattern;
```

➤ LIKE in Example:-

```
SELECT *  
FROM Customers  
WHERE country LIKE 'UK';
```